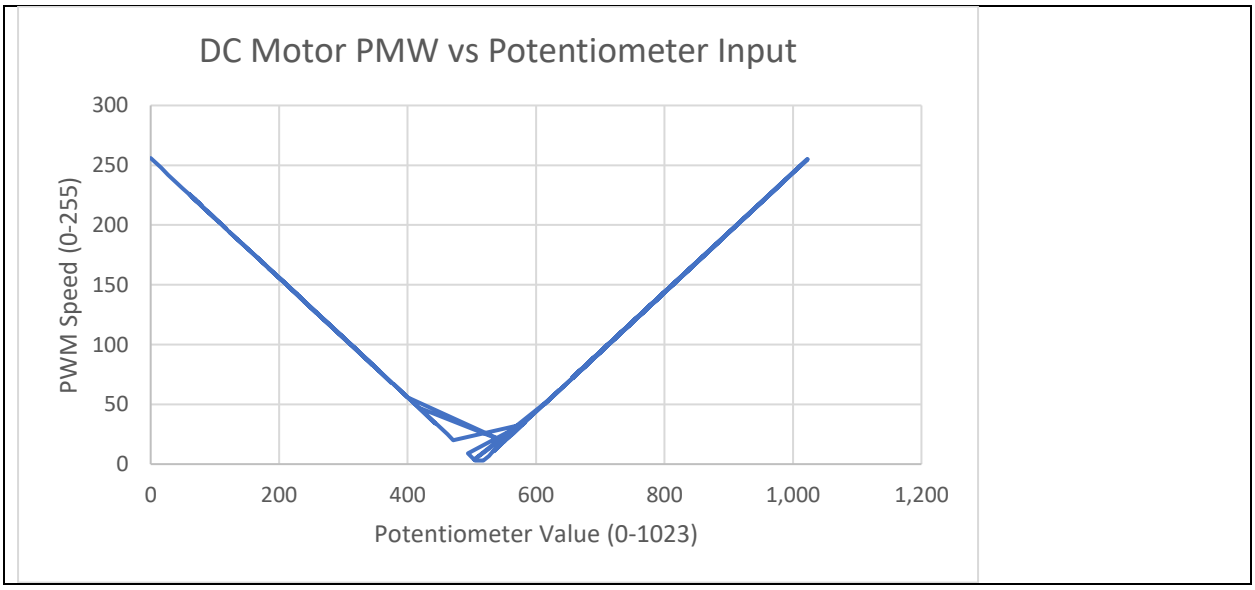
Lab 4 Report
Name: Tyler W. Grindrod
PID: 5695162 Date:

1. DC Motor. (40%) Extra 5% if external battery is used
    a. Create a code that controls a DC motor only using the basic PWM function analogWrite().
    b. The input needs to be an analog signal created by a potentiometer.
    c. Set the output to run the motor **backwards** for the bottom half of the potentiometer and **forwards** for the top half of the potentiometer.
    d. At either end of the input the motor should be running at full power.
        i. PWM output must be roughly 255 at both ends and 0 in the middle.
        1. PWM controls speed, setting the gates is HIGH or LOW controls direction.
        ii. PWM vs potentiometer input should be a linear relationship.
        iii. **Cannot write negative PWM**, will have issues.
        1. Map properly OR use Abs(<input>) returns absolute value of input field.
    e. Present a graph showing the input to output relation.
        i. X axis potentiometer reading, Y axis PWM value.
        ii. Add appropriate axes labels and title.
    f. Take a short video of the code running on your physical hardware.

Graph:



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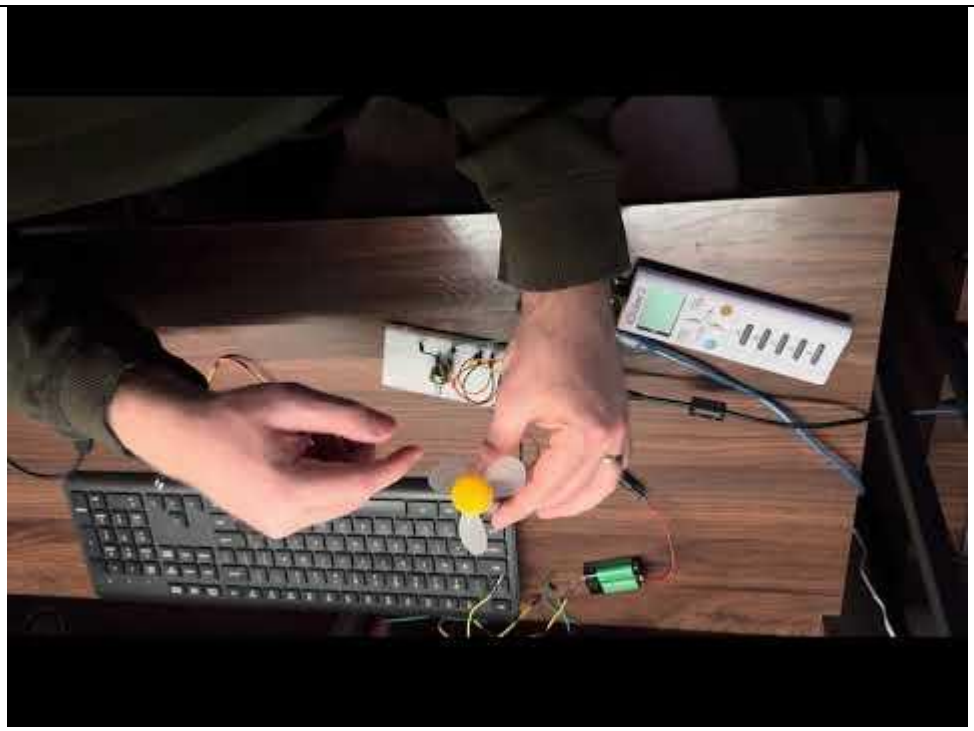
Video Link:



Code:

```
#define en12 9
#define gate1 8
#define gate2 11
// Potentiometer
#define POT A0

void setup() {
  pinMode(en12,OUTPUT);
// Gate1 & Gate2 helpe verse the Polarity of the stepper motor
  pinMode(gate1,OUTPUT);
  pinMode(gate2,OUTPUT);
```

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```
  Serial.begin(9600);
  delay(500);
  Serial.println("pot,pwm");

}

void loop(){
  int pot = analogRead(POT);
  int value = pot - 512;
  value = value/2;

  if (value > 0) {

    digitalWrite(gate1, HIGH);
    digitalWrite(gate2, LOW);
  }

  else if (value < 0) {
    digitalWrite(gate1, LOW);
    digitalWrite(gate2, HIGH);

  }

  else {
    digitalWrite(gate1, LOW);
    digitalWrite(gate2, LOW);
  }


  int speed = abs(value);
  analogWrite(en12, speed);
```

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```
  Serial.print(pot);
  Serial.print(",");
  Serial.println(speed);

  delay(50);

}
```

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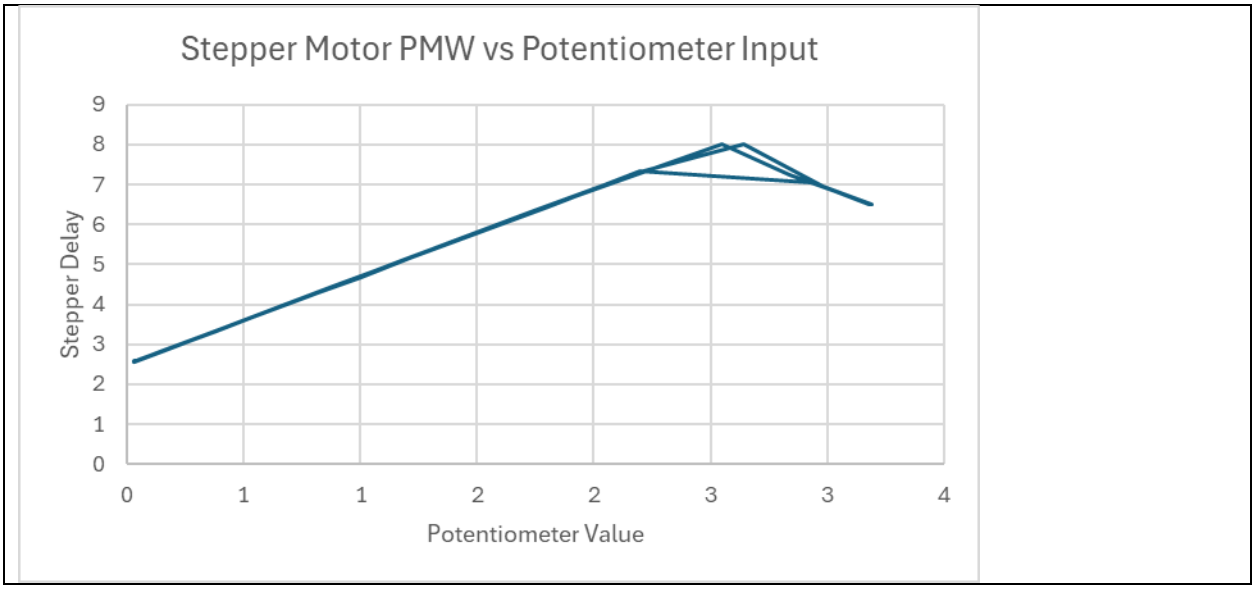
2. Stepper motor. (60%) Extra 5% if external battery is used
    a. Create a code that controls a stepper motor by setting the pins high and low with a variable time in between the steps using the delayMicroseconds() function.
    b. The input needs to be an analog signal created by a potentiometer.
    c. Set the output to run the motor **backwards** for the bottom half of the potentiometer and **forwards** for the top half of the potentiometer.
    d. **At either end of the input range the time delay should be the microDelay value**.
        i. Have small dead-band in center, maximum delay near center, minimum delay near edge.
        ii. The **delay should vary linearly with potentiometer input** going from max delay in the center to min delay at either extreme.
    e. Present a graph showing the input to output relation.
        i. X axis potentiometer reading, Y axis delay amount.
        ii. Add appropriate axes labels and title.
    f. Take a short video of the code running on your physical hardware.

Graph:



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Code:

```
#include <Stepper.h>

#define IN1 8
#define IN2 9
#define IN3 10
#define IN4 11

// Potentiometer
#define POT A0
unsigned long minDelay = 2500; // Fastest speed(2.5 ms)
unsigned  long maxDelay = 8000; // slow speed (8 ms)

int deadband = 30;
int stepNumber = 0;

void setup() {
  pinMode(IN1,OUTPUT);
  pinMode(IN2,OUTPUT);
  pinMode(IN3,OUTPUT);
  pinMode(IN4,OUTPUT);
  Serial.begin(115200);
}

void loop(){
  int potValue = analogRead(POT); // 0 to 1023

  float voltage  = potValue * (5 / 1023.0 );

  int distanceFromCenter = potValue - 512;
```

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```
unsigned long delayValue;


if (abs(distanceFromCenter) < deadband) {

  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
  delayValue = maxDelay;
}

else {
  int magnitude = abs(distanceFromCenter);
  // map from max Delay at center to mindelay at edges
  delayValue = map(magnitude,0,512,maxDelay,minDelay );
  if (distanceFromCenter > 0) {
    stepNumber++;
  }
  else {
    stepNumber--;
  }

    if (stepNumber > 3)stepNumber = 0;
    if (stepNumber < 0)stepNumber = 3;

    if (stepNumber == 0) {
      digitalWrite(IN1,HIGH);
      digitalWrite(IN2,LOW);
      digitalWrite(IN3,LOW);
```

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```
      digitalWrite(IN4,LOW);
    }

    else if(stepNumber == 1) {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, HIGH);
      digitalWrite(IN3, LOW);
      digitalWrite(IN4, LOW);
    }
      else if(stepNumber == 2) {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, LOW);
      digitalWrite(IN3, HIGH);
      digitalWrite(IN4, LOW);
    }

      else if(stepNumber == 3) {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, LOW);
      digitalWrite(IN3, LOW);
      digitalWrite(IN4, HIGH);

    }
  }

  delayMicroseconds(delayValue);

  static int counter = 0;
  counter++;

  if(counter > 200) {
```

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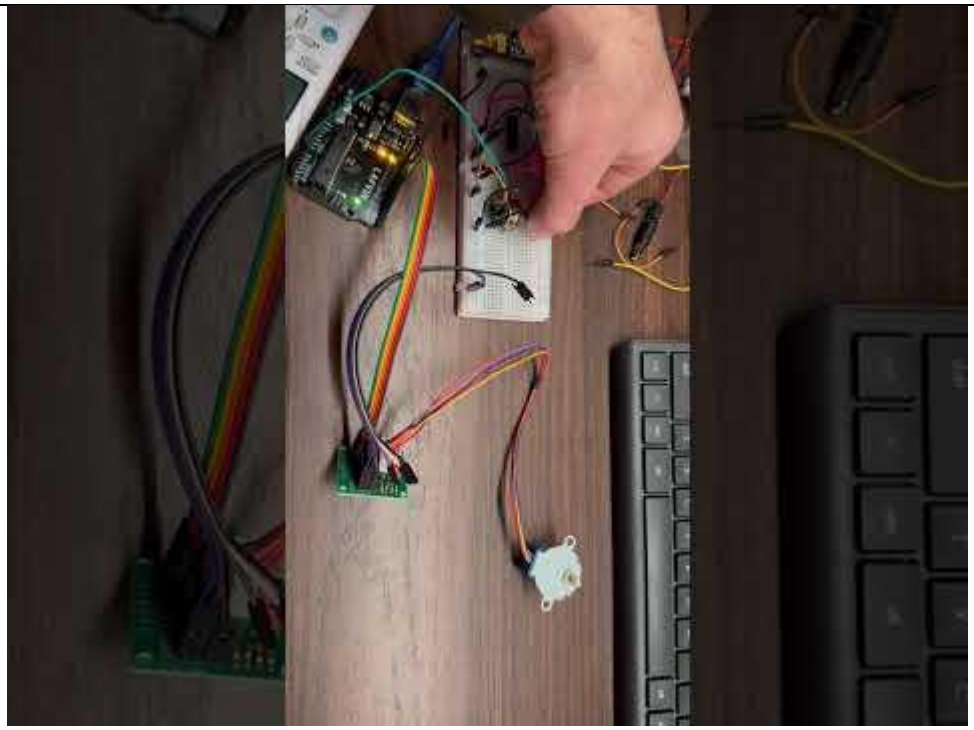
```
    Serial.print(voltage); // X-axis is input voltage
    Serial.print(";"); // semicolon seperator
    Serial.println(delayValue); // y-axis in milliseconds
    counter = 0 ;
  }

}
```

Video Link: 

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Objectives	Points	Full Marks	50% marks	25% marks
DC Motor.	40	1) Use of comments through the code explaining what is happening. 2) Motor can run forward and backwards with the same potentiometer. 3) Outputting max PWM at both extremes of potentiometer. 4) Graph contains proper labels for axis and title. 5) X axis is of the graph is the input voltage. 6) Video of code running on physical hardware.	1) Motor can run forward and backwards with the same potentiometer. 2) Graph contains proper labels for axis and title.	Motor can run forward and backwards with the same potentiometer.
Steppe Motor	60	1. Use of comments through the code explaining what is happening. 2. Motor can run forward and backwards with the same potentiometer. 3. Motor does not move in potentiometer center, has maximum delay just outside of the dead-band, and has minimum delay at both extremes of potentiometer. 4. Graph contains proper labels for axis and title. 5. X axis is of the graph is the input voltage. 6. Video of code running on physical hardware.	1. Motor can run forward and backwards with the same potentiometer. 2. Graph contains proper labels for axis and title.	Motor can run forward and backwards with the same potentiometer.